

msmetko.xyz
gitlab/msmetko
github/InCogNiTo124
linkedin/marijan-smetko
msmetko@msmetko.xyz
terra-incognita.blog

Zagreb, Croatia

About me:
Software engineer building and shipping production LLM features. I own the experimentation loop (evaluation datasets, A/B tests, and serving changes) and optimize for quality, latency, and cost.

Skills

Languages, tools & DevOps
Python, Bash, Rust, C/C++, Linux, Git, Jujutsu (jj), Buildkite, Kubernetes, Docker, Ansible, SQL

ML & data
PyTorch, JAX, NumPy, SciPy, pandas, scikit-learn, model evaluation, quantization

Public engagements

- Speaker at Web Summer Camp 2025: led a workshop on serving LLMs from first principles
- Python instructor: taught a six-week online Python course for FER students (2021)

Open source

- knarrow: knee/elbow finder
- chemeq: equation balancer
- Belot engine with an RL bot
- Jujutsu (jj) contributor
- Keplerian orbits visualizer

Beyond work

- Private pilot (EASA PPL(A))
- Self-hosts msmetko.xyz and side projects on Kubernetes
- Runs local LLMs on an eGPU rig
- Instruments: piano, accordion, recorder
- Big time science and space nerd

Marijan Smetko

Software Engineer · LLM Systems & Experimentation

Experience

July 2023 – present, *Software Engineer @ Google*

- Simplified production LLM serving for education features, cutting mean latency by 15% and p99 by 50% and freeing capacity worth thousands of TPU chips (20 SWE/y in cost)
- Improved answer accuracy for STEM homework solver on an education query slice by 25 percentage points offline (private evaluation set drawn from production traffic)
- Performed about 10 A/B experiments per quarter on Google Search (AI Overviews, AI Mode) and Lens, unblocking launches including Education Graphs in AI Overviews
- Turned a research notebook into a stable served visual-search demo, shown to external partners to demonstrate a capability that would become Find the Look and Circle to Search
- Fine-tuned Gemini-based embedding models into a binary classifier over multimodal web-search traffic

July 2021 – July 2023, *Junior AI Engineer* @ Photomath, now part of Google

- Implemented an internal platform that automated training orchestration, evaluation, and model deployment for ML experiments, adopted by the entire AI team (roughly 15 people)
- Built a model explainability and visualization tool that surfaced per-example failures and feature and region importance
- Designed and benchmarked ML models, including a handwriting-recognition inference system that reached 98%+ sequence-level accuracy

July 2020 – July 2021, *Computer Vision Intern* @ RealNetworks

- Built a model quantization pipeline for Google Coral TPU devices for edge computer vision
- Collected a dataset of thousands of images and trained a model for face anti-spoofing

April 2018 – September 2019, *Machine Learning Intern* @ Styria.AI

- Designed and implemented internal tools: request authorization service and a visual similarity model demo
- Analyzed deployed models and improved clustering accuracy of listings by more than 10 percentage points

Education

2019 – 2021, MSc in Computer Science, Faculty of Electrical Engineering and Computing (FER), University of Zagreb

- MSc thesis: *Incremental open-set recognition for semantic segmentation*

2016 – 2019, BSc in Computer Science, FER

- BSc thesis: *Visualization of class activation maps for image classification*